

Gaussian graphical models: just overview

Graphical models: continuation
Algebraic & Geometric Methods in Statistics

Sonja Petrović
Created for Math/Stat 561

Mar 25, 2026.

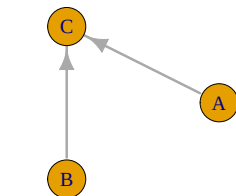
- Chapter 13: Graphical models
- after these short slides we will go over section 2.2 of Miles Bakenhus' course project lecture notes.

Examples

- Genes:
 - three genes in this example A,B,C
- Relationships:
 - A regulates C
 - B regulates C

BIOLOGY

- genes
- relationships



GRAPH

- vertices
- edges

$$P(A, B, C) = P(A)P(B)P(C|A, B)$$

PROBABILISTIC MODEL

- random variables
- statistical dependencies

Correlation vs causation

- Genes regulated as $X \rightarrow Y \rightarrow Z$
- Z and X are correlated, but do not interact directly

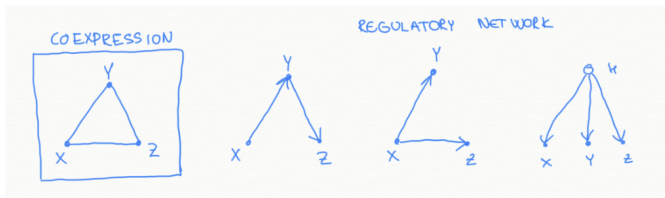


Figure 1: Source: K. Kubjas

Multivariate Gaussian random variables

- The CI statement $X_u \perp\!\!\!\perp X_v | X_{V \setminus \{u,v\}}$ is equivalent to the matrix $\Sigma_{V \setminus \{u\}, V \setminus \{v\}}$ having rank $|V \setminus \{u, v\}|$, or equivalently $\det(\Sigma_{V \setminus \{u\}, V \setminus \{v\}}) = 0$.
- This is equivalent to $(\Sigma^{-1})_{u,v} = 0$.
- The pairwise Markov property holds for a Gaussian distribution if and only if the entries of the concentration matrix corresponding to non-edges are zero.

Question (example)

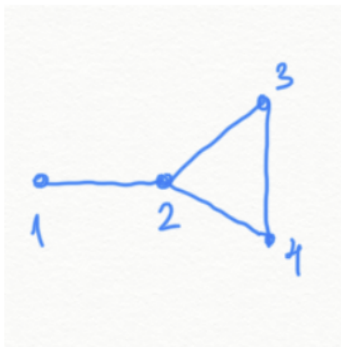
What is the form of the concentration matrices of a Gaussian distribution obeying the pairwise Markov property have?

1

$$\begin{bmatrix} k_{11} & 0 & k_{13} & k_{14} \\ 0 & k_{22} & 0 & 0 \\ k_{13} & 0 & k_{33} & 0 \\ k_{14} & 0 & 0 & k_{44} \end{bmatrix}$$

2

$$\begin{bmatrix} k_{11} & k_{12} & 0 & 0 \\ k_{12} & k_{22} & k_{23} & k_{24} \\ 0 & k_{23} & k_{33} & k_{34} \\ 0 & k_{24} & k_{34} & k_{44} \end{bmatrix}$$



Question (example)

What is the form of the concentration matrices of a Gaussian distribution obeying the pairwise Markov property have?

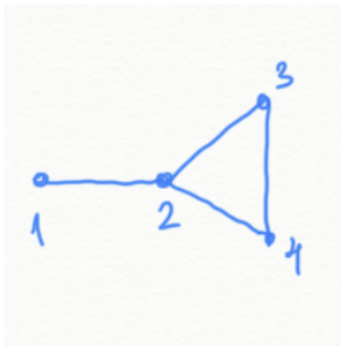
1

$$\begin{bmatrix} k_{11} & 0 & k_{13} & k_{14} \\ 0 & k_{22} & 0 & 0 \\ k_{13} & 0 & k_{33} & 0 \\ k_{14} & 0 & 0 & k_{44} \end{bmatrix}$$

2

$$\begin{bmatrix} k_{11} & k_{12} & 0 & 0 \\ k_{12} & k_{22} & k_{23} & k_{24} \\ 0 & k_{23} & k_{33} & k_{34} \\ 0 & k_{24} & k_{34} & k_{44} \end{bmatrix}$$

Correct answer: 2.



The usual... license

This document is created for Math/Stat 561, Spring 2023.

The slides that are not directly from the book are sourced from **Kaie Kubjas'** Algebraic Statistics course at Aalto University.

All materials posted on this page are licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](#).